



Fake and propaganda images detection using automated adaptive gaining sharing knowledge algorithm with DenseNet121

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Received: 12 March 2024 / Accepted: 24 June 2024

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Abstract

An additional tool for swaying public opinion on social media is to present recent developments in the creation of natural language. The term “Deep fake” originates from deep learning technology, which effortlessly and seamlessly steers someone toward digital media. Artificial Intelligence (AI) techniques are a crucial component of deep fakes. The generative powers of generative capabilities greatly reinforce the advancements in language modeling for content generation. Due to low-cost computing infrastructure, sophisticated tools, and readily available content on social media, deep fakes propagate misinformation and hoaxes. These technologies make it simple to produce misinformation that spreads fear and confusion to everyone. As such, distinguishing between authentic and fraudulent content can be challenging. This study presents a novel automated approach for the identification of deep fakes, based on Adaptive Gaining Sharing Knowledge (AGSK) and using DenseNet121 architecture. During pre-processing, the image’s sensitive data variance or noise is eliminated. Following that, CapsuleNet is used to extract the feature vectors. The deep fake is identified from the resulting feature vectors by an AGSK with DenseNet121 architecture, together with the hyper-parameter that has been optimized using the AGSK model. Propaganda and defamation pose less of a concern, and the results of the suggested deepfake image recognition model show how reliable and successful the model is. The accuracy of detection is 98% higher than other cutting-edge models.

Keywords CapsuleNet · DenseNet121 · Adaptive gaining sharing knowledge · Facial manipulation · Propaganda and deep fake images

1 Introduction

Since the invention of videography and imaging (Zhang et al. 2020), visual or audio modification has been practiced. Motion Graphics and image/video alteration depend heavily on the artifact frameworks for videos and photographs and their potent modifying features. Nevertheless, these artistic

creations can be quickly identified and distinguished using centered on human vision (Majumder and Kehtarnavaz 2020), therefore there is little room for their production. With the fast growth of technology and the use of processing methods such as countenance distortion an objective from the origin for a realistic look has appeared on an enormous level. This practice began with visuals, primarily for marketing and entertaining objectives. The question of whether facial emotions (Liu et al. 2021) are generic is brought up by the range of cultures in how they are represented.

The commonality of feelings among individuals in the manifestation of movements of the face continues to be one of the most contentious issues in psychology and cultural studies texts. DeepFakes are an exciting technique for changing realism that is becoming popular much more quickly than anticipated. A learning tool called DeepFakes (Yazdinejad et al. 2020) creates or alters audio and footage to portray individuals performing and expressing activities that they have never done or uttered. It could involve

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incorporating a face on an individual to make it appear as though the individual is doing a previously unheard-of action. It can change someone's true nature using either one of two techniques (Verdoliva 2020): either to fake their sounds or your photos and videos. Audio counterfeits are significantly easier to spot than Visible, the majority of currently used fakes are vision-based and are readily adjusted to have a genuine appearance.

Typically, it was only capable of making fun of performers and bureaucrats, as precision increases, fake information has started to propagate and cause a commotion among citizens by upsetting serenity and tranquility. The shifting of sentiments during a democratic process or implicating an individual in an offense is the current disturbing trend in the coverage of events. By synthesizing a replicated face switched by the motions and gestures of an additional individual, these alterations of the physical features (Stanciu and Ionescu 2021) of any significant representative, senator, or Executive are lifelike to the point that the average eye cannot differentiate between an authentic or fake movie. It may be mounted on the outermost part of a straight monitor panel. The modules, which are offered by electronic devices each comprise an image sensor and an autonomous central circuit that is linked by an adaptable thin wire.

Recently, fraudulent face data found in movies and pictures produced through computational alteration has drawn a lot of public attention. Since there aren't many advanced modifying instruments available, creating authentic fake face content requires a high level of knowledge and takes a long time, the amount and authenticity of computationally altered counterfeit facial components have been constrained. In contrast, over the last decade, academic disciplines of facial recognition (Serengil and Ozpinar 2020) and protection have paid increasing priority to comprehending and safeguarding against appearance attempts made against an individual's visage detector, which were thought to be a pertinent possibility to facial digital fingerprints. It enhances tissue healing soothes pain, enhances muscular performance, and makes the circulatory system flow more. If the program erroneously a certain individual, it can result in unfounded charges of theft or counterfeiting, the confidential nature of recognition of facial features (Verma and Verma 2020) technologies worries some consumers, and monitoring individuals without knowing it might be against personal freedoms. This following point summarizes the major contribution of this research;

- To remove noise and input image compression levels is analyzed using error level evaluation.
- To use DenseNet121 for the extraction of feature vector by means of both important blocks namely transition block and dense block.

- To compose AGSK algorithm with DenseNet121 architecture to detect the deep fake from the feature vector outputs.

The balance part of the paper is arranged like; Sect. 2 express the literature analysis of existing works based on deep fake images. The proposed methodology is proposed in Sect. 3 and the experimental investigations are analyzed in Sect. 4. At last, the paper is concluded in Sect. 5.

2 Literature survey

Zhu et al. (Zhu et al. 2020) have presented deep neural network (DNN) architecture identification of faces accomplished by combining an investigation process with Storage innovation and strengthened learning techniques. The three highest-performing systems are removed and retrained based on the premise offering. With pretty small system sizes, the system topologies people looked at were able to attain the highest levels of precision in the substantial facial database, achieving a precision of 98.77%. The effectiveness is demonstrated by the results it produces on these two well-known face collections. Hence, various loss functions are evaluated.

Ali et al. (Ali et al. 2020) have described an artificial neural network (ANN) dependent on ensemble classification for the evaluation of diverse facial expressions. For the automatic classification to categorize diverse facial features with greater precision, the emotion level location becomes apparent. By employing a culturally diverse collection for developing an assortment of binary classification techniques, it is possible to discover the differences in multilingual language interpretation. Improved the classifier's capacity to correctly foresee the occurrence of a statement. Therefore, the interconnected face emotion identification is quite uneven.

Wang et al. (Wang et al. 2020) have modified the investigation on Generative Adversarial Networks (GAN) and analyzed multiple uses in the area of GAN-based picture generation. Additionally, the outlined techniques applied in deployments enhanced the output quality of produced pictures. Individuals offered various approaches to instructing and assessing the method since, even though the investigation is getting increasingly developed, it continues to encounter certain obstacles, including inconsistent learning and difficult evaluation. Thus, it is not suitable for the development of other intriguing applications.

Pan et al. (Pan et al. 2020) have developed deepfake detection technologies a combination of two cutting-edge neural systems and four approaches for creating fake videos that were tested and contrasted. The four prototypes were combined and used through a selection system to identify

every single group of films produced by each of the common counterfeit techniques. This used a straightforward approach in which any video labeled as fraudulent by any means was finally categorized as fake. The algorithm fared more successfully at identifying actual videos as opposed to false pictures, with a 2–3% boost in reliability. Thus, inter-frame design associations weren't taken into account.

Hashmi et al. (Hashmi et al. 2020) have implemented a Convolutional Neural Network (CNN) pathway for the immediate identification of contrasts in incredibly small perceptual examines on the faces of actual and false frames. The representation was trained using a collection of pictures, both real and fraudulent, that were gathered from various sources. The suggested algorithm and created network establish a new standard for identifying visual forgeries and demonstrate how this system can produce results that are relevant to any phony-produced video or picture. The recommendation may accurately and effortlessly recognize visual fraud in movies and photos. However, the framework is computationally intensive for its weight class.

Chintha et al. (Chintha et al. 2020) have evaluated convolutional recurrent neural networks to find deep fakes as a clear and effective portrayal of face features. To assist in identifying periodic discrepancies, these images are transmitted into bilateral repetition stages. These methods integrate bilateral repetitive frameworks, entropy-based cost estimates, and multilayer implicit forms. To derive content from the transcriptions that are conceptually vibrant, the implicit structures for video as well as noise are meticulously determined. The resistance achieved by the techniques to enlargement is the absence of prediction for voice fake identification. Therefore, it was less successful on the bigger datasets.

Yang et al. (Yang et al. 2021) suggested an intelligence forensic method to improve the directed characteristics, or pattern objects, that were included in the face alteration

picture. Even though the majority of this distortion is indistinguishable in appearance, the Resnet18 system, which can continually decrease the sample size and resample, will eventually discover how to distinguish between increased appearance shifts and actual and fraudulent facial pictures. It can obtain the best precision for identification in both complete picture learning and facial development. Hence, it is inadequate to develop an initial verification system.

Xu et al. (Xu et al. 2021) highlighted a set convolutional neural network (SCNN) to address face-altered video recognition from a particular position to combine multi-frame characteristics to recognize facial distortion movies. Features on numerous footage frames are regarded as a set that represents the procedures used to create the facially modified film. The map representations depicting the associated faces are produced in this collection because the identical underlying system is utilized to calculate characteristic images for every single face in the video image. The effectiveness was demonstrated by training and testing it on the datasets. Thus, it is not suitable for creating set reduction structures.

3 Proposed framework

The proposed deep fake identification model with the workflow diagram is discussed in this section. Figure 1 employs the proposed overall structure and the core phases are pre-processing, feature vector extraction and deepfake detection model. The following subsection delineates each core phases.

3.1 Pre-processing

When the image tend to digital editing, the error level evaluation (ELE) measuring the varying compression levels of

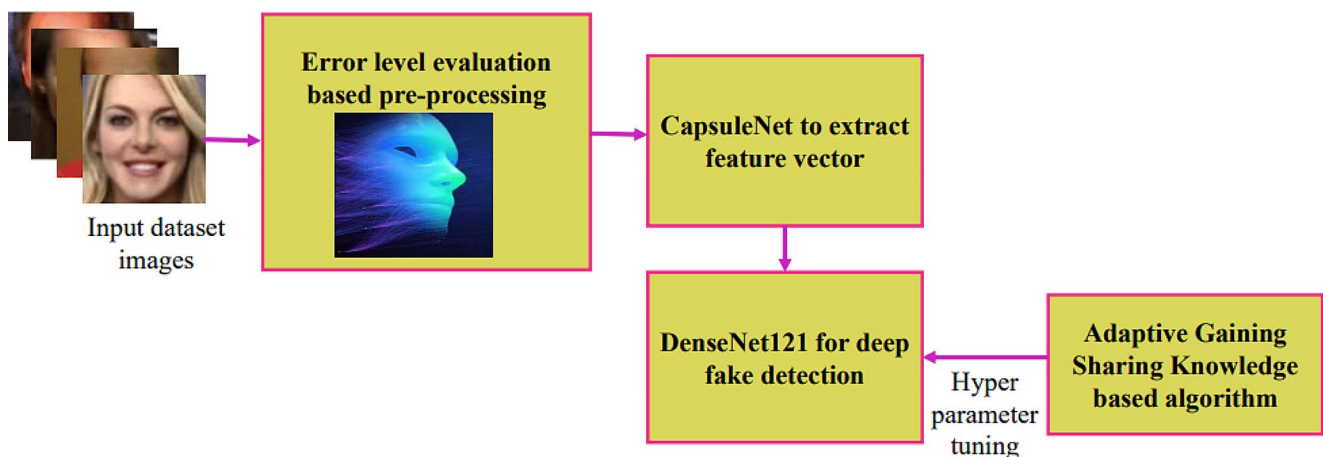


Fig. 1 Proposed workflow structure

an image (Rafique et al. 2023). Use the compression level to the acceptable compromise among the quality and size of an image. Where, 10:1 is the ratio of JPEG compression and freely compress the 8×8 pixel image grids with JPEG technique. There is a lack sufficient information when any matrices less than 8×8 and it demonstrates poor quality of image compression. While allow the image resaving, each 8×8 grids to similar level of error. At the same pace, roughly deteriorated each square and the faults uniformly distributed to an image (Mostafiz et al. 2018). Compared to the rest images, higher error potential change grid in an improved image. Computed the variation among both images and the error rate of 95% resave the image. If the pixels in local minima, check any variation in cell are determined using ELE and in the database, any digital tampering are determined. Also, the image noises are removed.

3.2 Feature extraction

CapsuleNet extract the relevant feature vectors of deep fake face images from pre-processed data. Compared to CNN, the positional properties like relations and orientation among features are recorded using CapsuleNet and demonstrating higher accuracy during deep fake detection. In the loopholes, CNN filled a Capsule networks (Ryu and Jang 2022). Because of ineffective pooling functions, a spatial information are obtained with the difficulties of CNNs. CapsuleNet utilizes dynamic routing model. It is the combination of neurons or capsules collections, which is independent in nature. Vectors are the CapsuleNet output in which the softmax pooling are not utilized. Utilizes the function of squashing and the capsules vector input and output are computed. Following formula outline the function of nonlinear squashing.

$$T = \frac{\|V_R\|^2}{1 + \|V_R\|^2} \times \frac{V_R}{\|V_R\|} \quad (1)$$

The capsule input and output is R and $\|V_R\|$. In the vector model, the capsule output is T . Depending upon the capsule decision, the routes of single capsule output to extra capsule and agreement protocol routing is utilized (Rajalakshmi and Senthilkumar 2023). High capsules order utilizes output from low capsules. From any feature loss, widely escapes from this because of the avoidance of pooling and the time is saved. Effectively address the Picasso issue and better generalization power and it effectively extract the feature vectors from deep fake face images and the dimensionality of features are reduced then it fed for deep fake detection process. Figure 2 depicts the structure of CapsuleNet.

3.3 Deep fake detection

This section comprises the deep fake detection model using an automated adaptive gaining sharing knowledge (AGSK) based algorithm with DenseNet121 architecture.

3.3.1 DenseNet121

In a feed-forward fashion, one of the CNN model is DenseNets in which every layers are linked with each other. Each previous layer with its feature maps are concatenated in the input layer. Where, LA is the layer in the network. Based on the subsequent $LA-1$ layer, fed the resulted feature map and feature map of each existing layers are combined with the input I and $\frac{LA \times (LA+1)}{2}$ is total connection among the layers. While contrasted with conventional CNN, few parameters only required here and it not required to relearn the feature map redundancy. Based on deep training

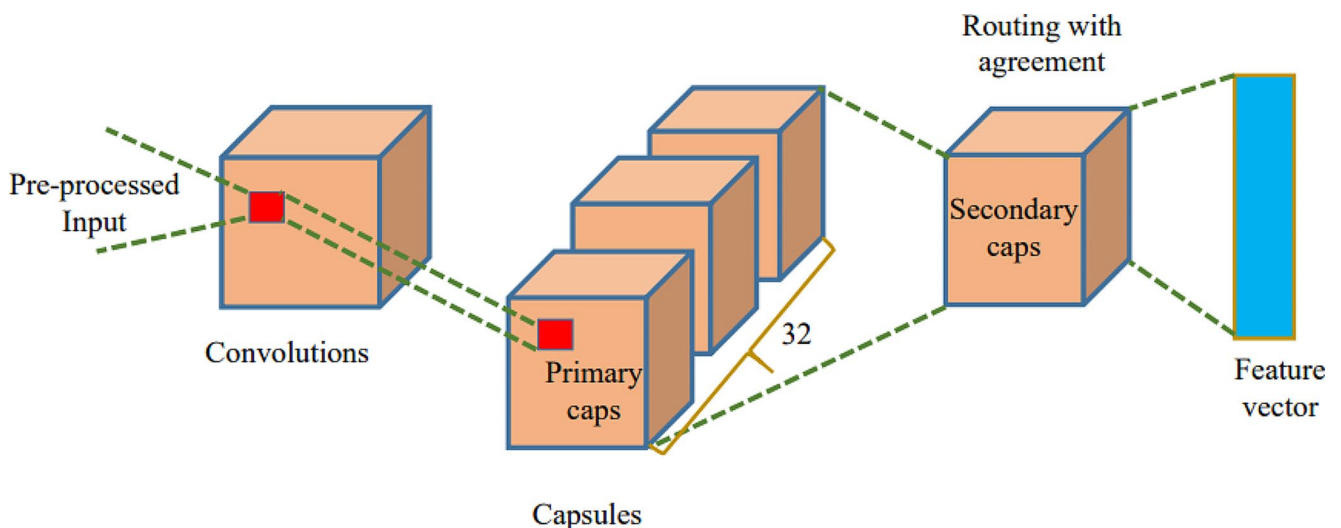


Fig. 2 Outline structure of CapsuleNet

structures, the vanishing gradient issues are minimized due to the pattern of dense connection (Ezzat et al. 2020). From the loss function, every layer directly accessing to gradients that pass the data to network as well as gradient flow is enhanced.

Here, I as the layer index with $NH_I(\cdot)$ represent non-linear transformation and LA as the layer of DensNet (Solano-Rojas et al. 2020). The convolution (Conv) and ReLU follows the batch normalization function. In the network, below formula express the input layer's (LA) dense connectivity.

$$Y_I = NH_I([Y_0, Y_1, \dots, Y_{I-1}]) \quad (2)$$

From 0 to $LA-1$ layers, the feature map concatenation is $[Y_0, Y_1, \dots, Y_{I-1}]$ with the input layer is Y_I . Both transition block and dense block are the two major blocks involved in this network. Every layer of dense comprises 3×3 Conv and 1×1 Conv layers respectively. The block of transition involves 2×2 average pooling, 1×1 Conv and batch normalization layers. Four dense blocks with DenseNet network is named as DenseNet121 that contains a sequence of 6, 12, 16 and 24 dense blocks and Fig. 3 represents the structure of DenseNet121.

3.3.2 Adaptive gaining sharing knowledge (AGSK) algorithm

This section is used to elucidate the AGSK algorithm for tuning the hyperparameters of DenseNet121 technique while effectuating the fake image detection from the given dataset. First the Gain Sharing knowledge (GSK) algorithm is explained and followed by adding the chaotic map for the improvisation of adopted GSK algorithm. This algorithm relies on the concept of sharing and obtaining the knowledges by the human beings in their life and employs two stages stated below,

3.3.2.1 Junior gaining and sharing phase The knowledge gaining is obtained from the family members, closed ones or relatives by this phase. The knowledge gaining is in small network since not having that much exposure to the world (Mohamed et al. 2021).

3.3.2.2 Senior gaining and sharing phase These phase people gain knowledge from the friends, social circles, and colleagues and shar their knowledge to that circle and thus improvise their knowledge. This phase people easily can differentiate the good and bad people.

The numerical expression for GSK is delineated in this section. Let us assume, y_{ij} ; $i = 1, 2, \dots, M_p$; $j = 1, 2, \dots, D$ as the individual population or features from the total population. The total aggregated population is M_p and the disciplines are D . then the fitness values can be evaluated as, $F(y_{ij})$; $i = 1, 2, \dots, M_p$; $j = 1, 2, \dots, D$ for the individual y_{ij} .

The random solutions for the population are initialized as follows,

$$y_{ij}^0 = R_j + C_j * (UP_j - LP_j) \quad (3)$$

The random integer C is uniformly distributed over the range of 0 and 1 with the lower and upper constraints of the optimization issue are LP and UP . The locations of junior and senior phases are done with the evaluation of dimensionality as shown below,

$$D_K = \text{ceil} \left(D \times \left(1 - \frac{mE}{NmE} \right)^L \right) \quad (4)$$

$$D_S = D - D_K \quad (5)$$

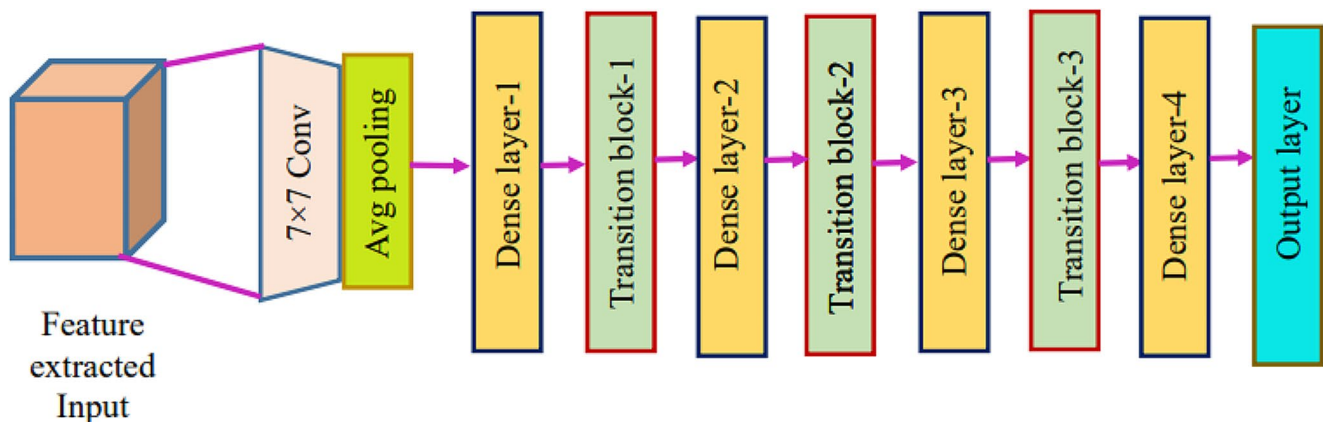


Fig. 3 General structure of DenseNet121

The dimensionalities of the junior and senior gaining and sharing phases are implied as D_K and D_S correspondingly. The function estimations and the maximum number of function estimations are implied as mE and NmE correspondingly. The knowledge rate is implied as L and is a real number with positive value. This number can be used to manage the experience rate of every person throughout the generations.

The probability of the junior gaining and sharing phase is evaluated and upgraded by organizing the individuals based on the ascending order and found the objective function numeric,

$$y_B, \dots, y_{i-1}, y_i, y_{i+1}, \dots, y_W \quad (6)$$

Select the adjacent best and worst value for each respective $y_i (i = 1, 2, \dots, M_P)$ as y_{i-1} and y_{i+1} . The senior gaining and sharing phase relies on the impacts of other people on their life and hence we follow the following procedure for the updates,

The individuals are grouped into best, better, and worst based on the objective function numeric in an increasing order. $Best\ Individual = 100P\%(y_B)$, $Middle\ Individual = D - 2 \cdot 100P\%(y_M)$, $Worst\ Individual = 100P\%(y_W)$.

3.4 Probability estimator of junior phase

The probability evaluation operator $\Pr(GJ_{ij})$ can be expressed as,

$$\Pr(GJ_{ij}) = 1 / \left[1 + \exp \left[-2W * \left(By_{ij}^{New Jr} - 0.5 \right) / (1 + 2 * j_F) \right] \right] \quad (7)$$

The knowledge factor of the algorithm is j_F with positive constant value and is evaluated for each individual By_{ij} . The bandwidth with constant and positive value is W (Mohamed et al. 2021). The respective binary individual value can be estimated as

$$By_{ij}^{New Jr} = \begin{cases} By_i + j_F(q) * [(By_{i-1} - By_{i+1}) + (By_r - By_i)]; & \text{if } F(By_i) > F(By_t) \\ By_i + j_F(q) * [(By_{i-1} - By_{i+1}) + (By_i - By_r)]; & \text{otherwise} \end{cases} \quad (11)$$

The adaptive GSK for senior phase can be expressed as,

$$By_{ij}^{New Sr} = \begin{cases} By_i + j_F(h) * [(By_{PB} - By_{PW}) + (By_{Middle} - By_i)]; & \text{if } F(By_i) > F(By_{mid}) \\ By_i + j_F(h) * [(By_{PB} - By_{PW}) + (By_i - By_{mid})]; & \text{if } F(By_i) = F(By_{mid}) \\ By_{ij}^{old}, & \text{otherwise} \end{cases} \quad (12)$$

$$Bv_{ij}^{Jr} = \begin{cases} 1, & \text{if } C \leq \Pr(GJ_{ij}) \\ 0, & \text{if } \text{Otherwise} \end{cases} \quad (8)$$

The probability vector of junior phase of the j^{th} element is $\Pr(GJ_{ij})$.

3.5 Probability estimator of senior phase

The probability of the senior phase is evaluated as below,

$$\Pr(GS_{ij}) = 1 / \left[1 + \exp \left[-2W * \left(By_{ij}^{New S} - 0.5 \right) / (1 + 2 * j_r) \right] \right] \quad (9)$$

The selection of bandwidth factor for our solution is predominant and it can be expressed as,

$$Bv_{ij}^S = \begin{cases} 1, & \text{if } C \leq \Pr(GS_{ij}) \\ 0, & \text{if } \text{Otherwise} \end{cases} \quad (10)$$

The probability vector of the senior phase is given as $\Pr(GS_{ij})$.

3.6 AGSK model

To improvise the GSK we have included chaotic map based sequence to enhance the parameters of GSK. For that we have taken two factors such as knowledge factor (j_F) and knowledge ratio (j_r). These two factors are selected arbitrarily to manage the shared and gained knowledge among the people. To improvise the GSK algorithm these factors are replaced with chaotic sequence and renamed the algorithm as Adaptive Gaining and sharing knowledge based algorithm (AGSK). This addition will avoid the GSK falling for local optimization and search for global optima. This also balances the two aspects of the algorithm. The adaptive GSK for junior phase can be expressed as,

3.6.1 An AGSK algorithm with DenseNet121 for deep fake detection

The final detection process and training procedure is affected due to structural and non-structured hyper parameters tuning of DenseNet121. Hence, the AGSK model adopts to optimize the layers of DenseNet121 also tuning its hyper-parameters. Based on few rules of network encoding, the hyper parameters are tuned effectively by means of best solution in AGSK. The step involved for deep fake detection model by means of AGSK algorithm with DenseNet121 model is elucidated as below;

- Initialize the DenseNet and proposed AGSK parameters along with the binary population for the detection of deepfake images
- Evaluate the fitness value of each feature from the images of deepfake images
- The global best solution is chosen from the binary population
- The dimensionality of the Junior GSK and senior GSK can be evaluated using the eqns. (4) and (5)
- The location updates of the individuals are effectuated with the Eq. (7) for the junior GSK
- The location updates of the individuals are effectuated with the Eq. (9) for the senior GSK
- Appending the Chaotic sequence for the enhancement of GSK
- Update the location based on the eqns. (11) and (12) for the population of both phases
- Detection of best solution
- If the stopping condition occurs return the detected result

4 Simulation results

This section extends the simulation results of proposed work to detect the deepfake images from the real time applications. This section presents the dataset used and parametric evaluation and comparative study of proposed with state-of-art works such as DNN (Zhu et al. 2020), ANN (Ali et al. 2020), GAN (Wang et al. 2020) and CNN (Hashmi et al. 2020).

4.1 Simulation setup

The simulation is demonstrated in a system with the running capacity of windows 10 7th generation, i7 Intel core CPU. For simulation Python 3.8 has been adopted and the details are enlisted in Table 1.

Table 1 Simulation setup

Components	Description
RAM	8 GB
OS	Windows 10
CPU	Intel core i7, with 2.8 GHz processor
Language	Python 3.8
GPU	Nvidia

4.2 Dataset description

For the performance evaluation we have taken the publicly available dataset known as Yonsei University's Computational Intelligence and Photography Lab (YUCIPL) (Real and Fake Face Detection 2019). This dataset contains expert-generated high-quality photoshopped face images. The images are composite of different faces, separated by eyes, nose, mouth, or whole face. It contains real/fake face photos, respectively. In case of fake photos, it has three groups; easy, mid, and hard. It contains both the fake and real human faces and is developed in order to detect the deepfake images from the video or anyother applications. The dataset provides details about fake and real images incorporated with the personal details such as ethnicity, age, sex, and the technique used for the manipulation of fake images. The sample real and fake images from the dataset are shown in Table 2. The parameter settings used for the proposed work is depicted in Table 3.

4.3 Metrics used for the evaluation

The metrics used for the evaluation validation of proposed and state-of-art works are elucidated in this section. We have taken the accuracy, specificity, and sensitivity of the metrics to validate the performances. The definitions are expressed as below,

4.3.1 Accuracy

It is measured as how close the agreement of the outcome and the actual value of the thing that has been measured.

$$A = \frac{TN + TP}{TP + TN + FP + FN}$$

Here TN is the true negative of the dataset measured, TP is the true positive of the measure dataset, FP is the false positive of the dataset, and FN is the false negative of the dataset.

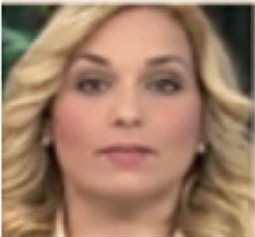
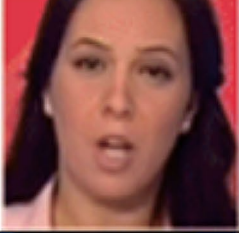
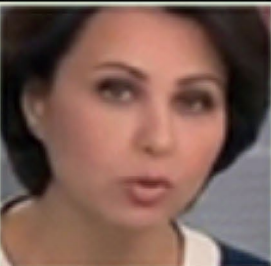
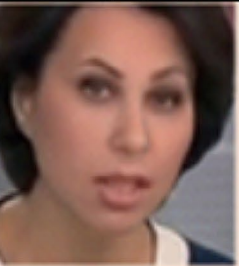
Real Images	Deepfake Images
	
	
	
	
	
	
	

Table 2 Sample images of both real and deepfake images

Table 3 The parameter settings of proposed approach in the detection of deepfake images

Factors	Ranges
Knowledge rate	10
M_p ($D < 20$)	60
M_p ($D > 20$)	150
NmE ($D < 20$)	6000
NmE ($D > 20$)	25,000
W (Bandwidth factor)	7
Pr (probability)	0.1
Number of iterations	100

4.3.2 Specificity

Specificity is measured as how correctly the subjects are detected as deepfake from the available deepfake images and can be expressed as,

$$SP = \frac{TN}{FP + TN}$$

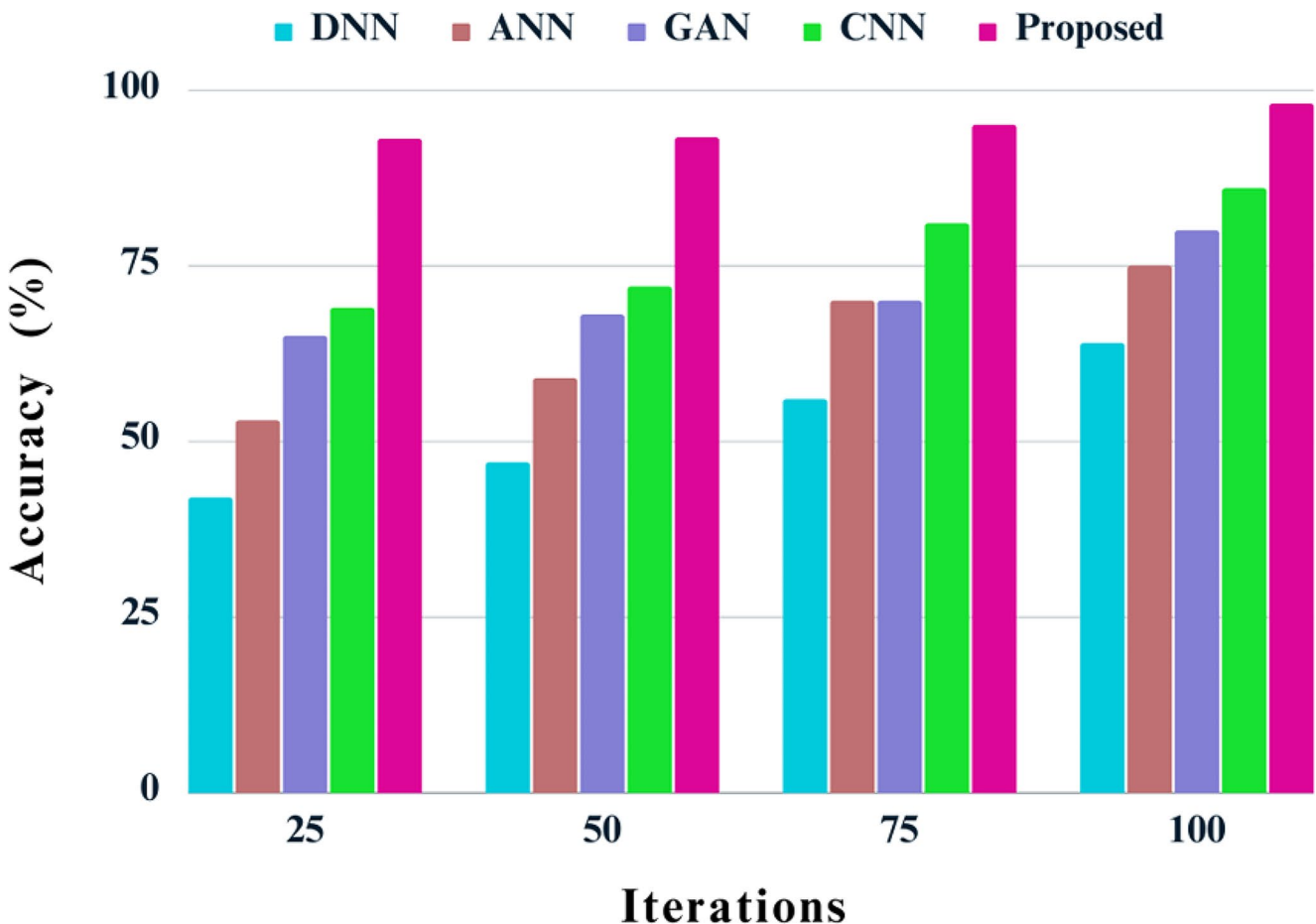
4.3.3 Sensitivity

It can be measured as how correctly the subjects are detected as real images from the available real images and can be expressed as,

$$SN = \frac{TP}{FN + TP}$$

4.4 Performance assessment

The optimization model based DenseNet structure is to detect the deepfake. The state-of-art result for accuracy is visualized in Fig. 4. The evaluation of accuracy is computed by means of DNN, ANN, GAN and CNN methods. The accuracy of 64%, 75%, 80%, 86% and 98% is undertaken by using DNN, ANN, GAN and CNN and proposed methods at 100th iteration. Based on the existing methods like DNN, ANN, GAN and CNN, this graphical representation clearly states that the superior accuracy of 98% during deep fake image identification. The proposed work overtake

**Fig. 4** State-of-art investigation for accuracy

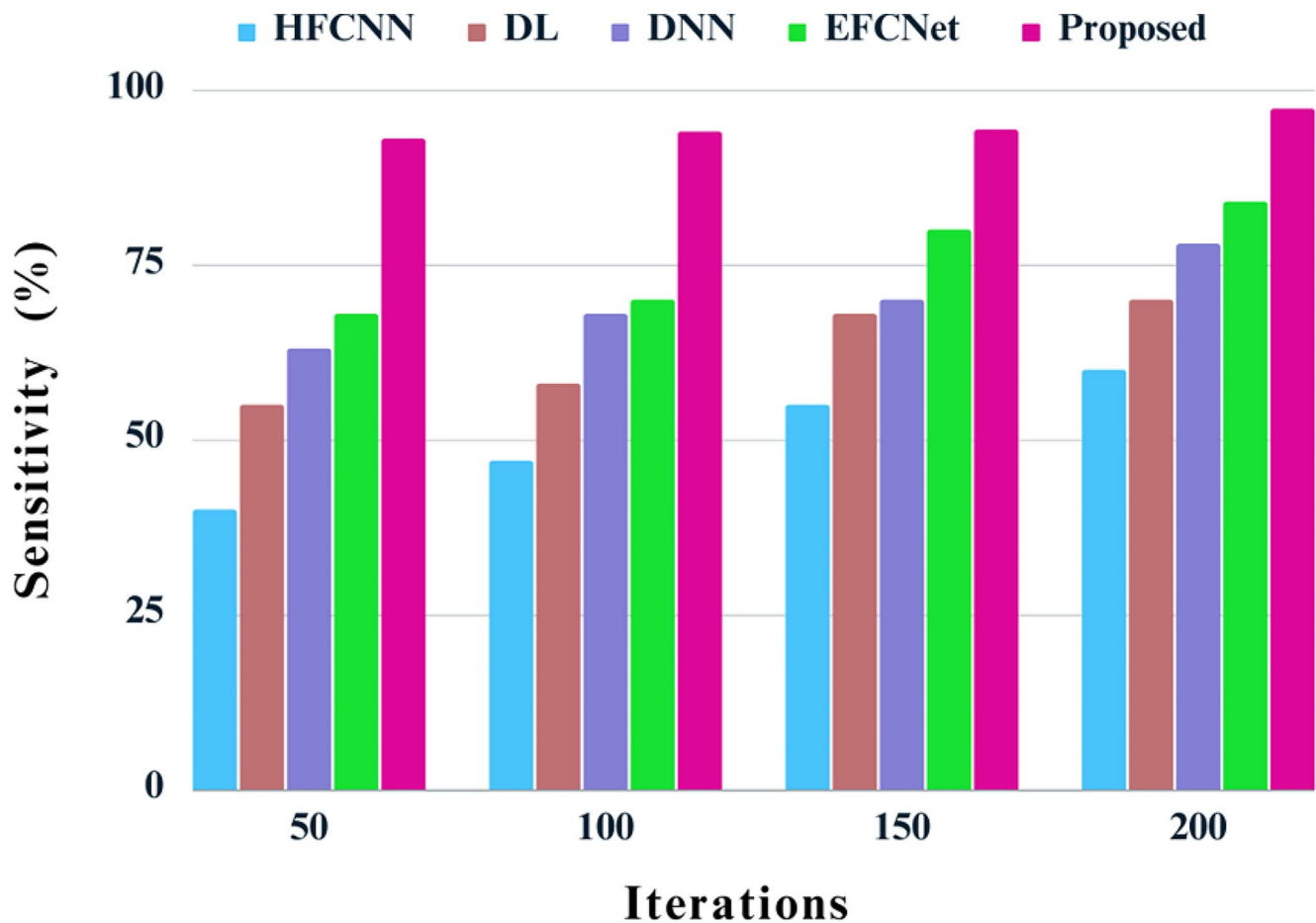


Fig. 5 State-of-art investigation for sensitivity

better accuracy rate with minimal error values to identify deep fake images, which observed a better results than other previous methods.

The state-of-art comparison of sensitivity outputs are displayed in Fig. 5. Change the number of iterations from 25 to 100 based on various percentages of sensitivity. The methods namely DNN, ANN, GAN and CNN are used to calculate the sensitivity percentages of deep fake image identification. At 100th iteration, the methods such as DNN, ANN, GAN and CNN and proposed undertakes the sensitivity percentages of 60%, 70%, 78%, 84% and 97.3% respectively. The proposed method delivers the highest sensitivity of 97.3% comparing with previous methods namely DNN, ANN, GAN and CNN. Compared with these previous models, the proposed sensitivity results increases by increasing the number of iterations.

Figure 6 outlines the graphical representation of specificity results. The deep fake identification methodologies like DNN, ANN, GAN and CNN and proposed are selected for this graphical plot. The iteration increases by varying the specificity percentages. By comparing this results, these methods DNN, ANN, GAN and CNN and proposed offers

65%, 74%, 79%, 85% and 96.43% at the level of 100th iteration. The proposed method delivers the superior specificity of 96.43% comparing with previous methods namely DNN, ANN, GAN and CNN. Compared with these previous models, the proposed specificity results increases by increasing the number of iterations. The proposed work demonstrates superior rate of specificity to identify deep fake images, which observed a better results than other state-of-art approaches.

Figure 7 displays the comparative plot for computation time. The computation time during deep fake image identification is measured in terms of milliseconds (ms). This plot undertakes the computation time of 82 ms for DNN, 60 ms for ANN, 42 ms for GAN, 51 ms CNN and 18 ms for proposed work. These investigation reveals the computation time of proposed work is minimum to DNN, ANN, GAN and CNN.

The suggested model has many issues and limits, according to the authors, including the fact that it is computationally costly and necessitates the employment of sophisticated and unique processing units. It takes a long time to train for complicated jobs on a subpar CPU. They rely on the

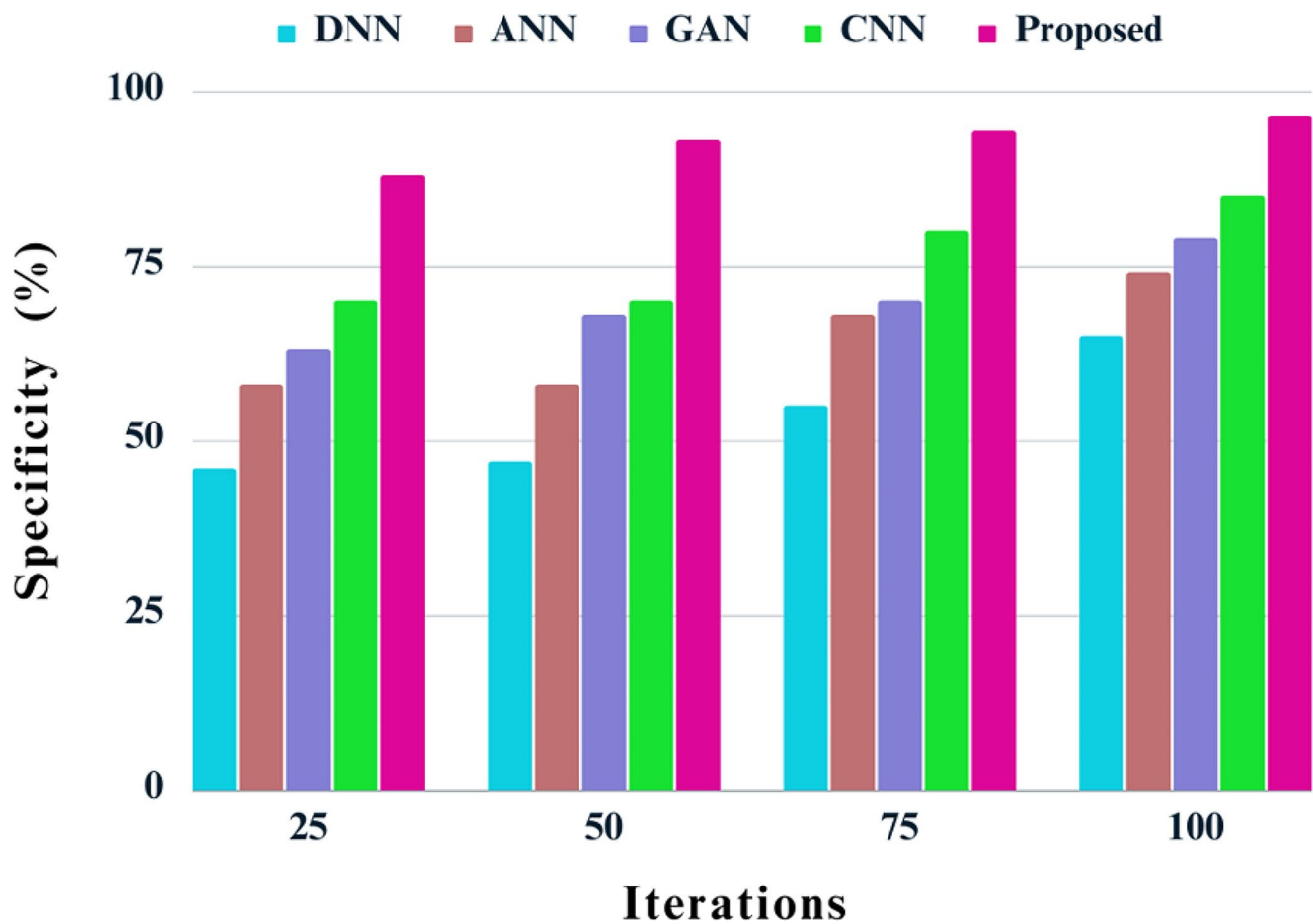


Fig. 6 State-of-art investigation for specificity

quantity of data that is given to them, which is another issue. Poor performance may be expected if there is little data, and vice versa. Millions of parameters in the suggested model need a vast quantity of data. As a result, using the suggested model necessitates a lot of training data and takes time to train.

5 Conclusion

The task in this article is to use the suggested DenseNet based AGSK optimization strategy to distinguish between real and deepfake photos. To do this, we pre-processed the photos from the YUCIPL dataset. From now on, the CapsuleNet technique is used to extract the features that are used for the detection of deepfake images. The pre-processing technique was used for sensitive data persistence and noise removal. Using the training and testing dataset, the Densenet approach distinguishes the deepfake images and real photos from the dataset with ease. However, low or maximum data availability will lead to overfitting errors; to address this problem, the suggested AGSK technique has been applied.

This method efficiently adjusts Densenet's hyperparameters. Additionally, models were run and the performance analysis of suggested and cutting-edge works like CNN, ANN, GAN, and DNN was examined. With an accuracy of 98% and a computation time of 18 ms, our work is resilient and improves the identification of deepfake images. Deep learning models require large datasets of high-quality, labeled images to train effectively.

Obtaining such datasets, especially with accurately labeled fake and propaganda images, can be challenging. Propaganda often relies on context and narrative, which can be challenging for models to grasp purely based on visual data. By focusing on Improved Dataset Collection and Annotation, Robust Model Architectures and Real-Time Detection Systems, future research can enhance the capabilities of deep learning models in detecting fake and propaganda images, making them more robust, accurate, and usable in real-world scenarios.

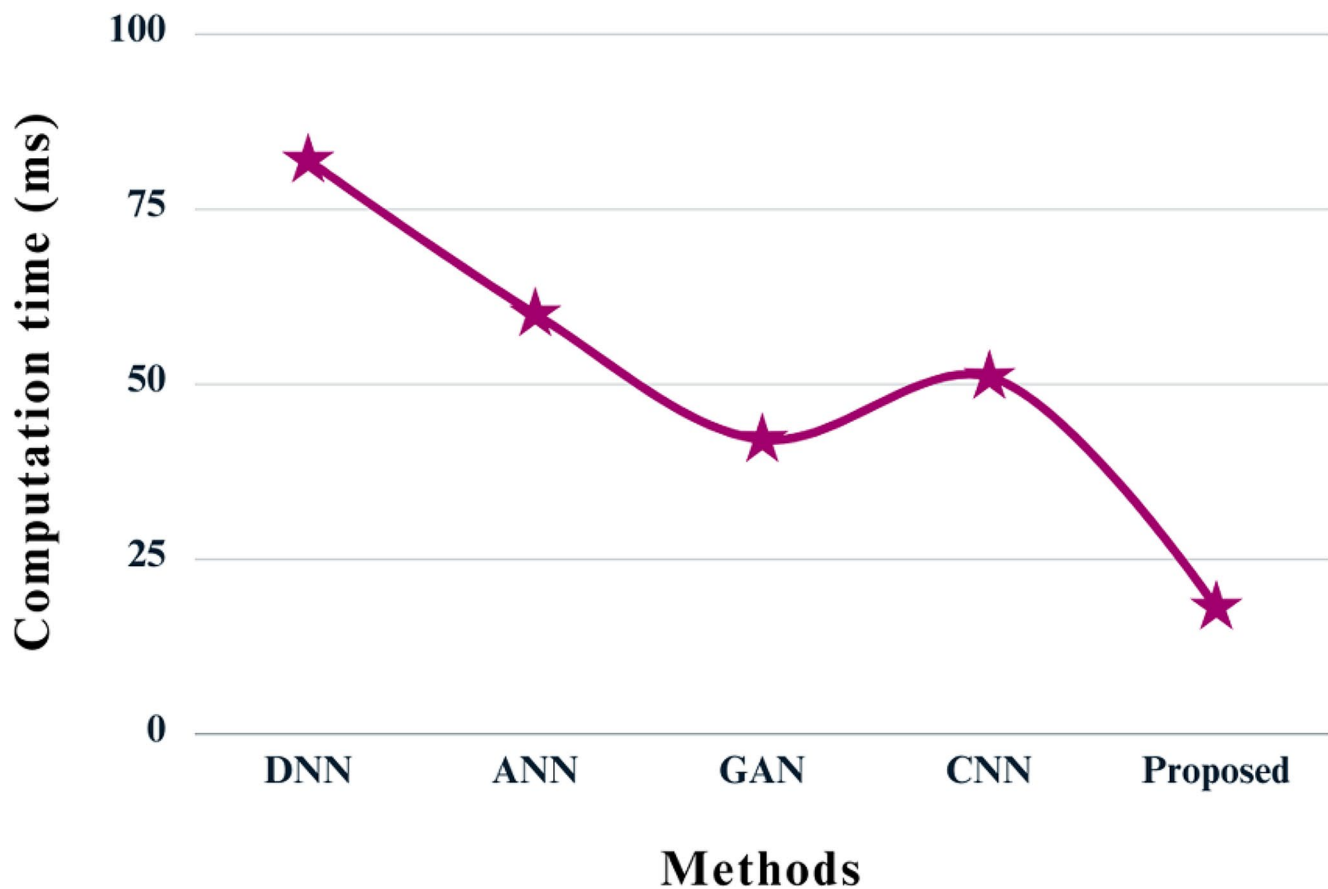


Fig. 7 State-of-art investigation for computation time

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